

ECE 3337: Signals & Systems Analysis

**Lecture 27: Fourier → Sampling Theorem
(LAST LECTURE)**

**Reference: Chapter 6
(Section 6.26)**



Important Reminder

- Please remember to complete your online course evaluation on <https://accessUH.uh.edu>



Faculty/Course
Evaluation

- Your candid feedback will help us further improve ECE 3337



Final Exam

Date: May 11th 2 - 4pm, SW 102 (Social Work Building).

<https://www.uh.edu/infotech/services/facilities-equipment/supported-classrooms/sw/102/>

Plenty of old exams are posted on MS Teams for practice

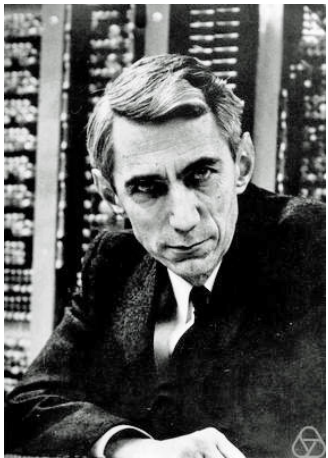
- Practice for accuracy and speed
- Test yourself individually, rather than as a group
- Avoid a false sense of knowing a topic
 - This can easily happen in a group setting
- Suggested study strategy:
 - Do the practice tests individually
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 - Discuss the challenging problems in your study group
 - Follow up with individual work to confirm your understanding



Recap: The Sampling Theorem



Harry Nyquist
(1889 - 1976,
Swedish American, Texan)



Claude Elwood Shannon
(1916 – 2001, USA)

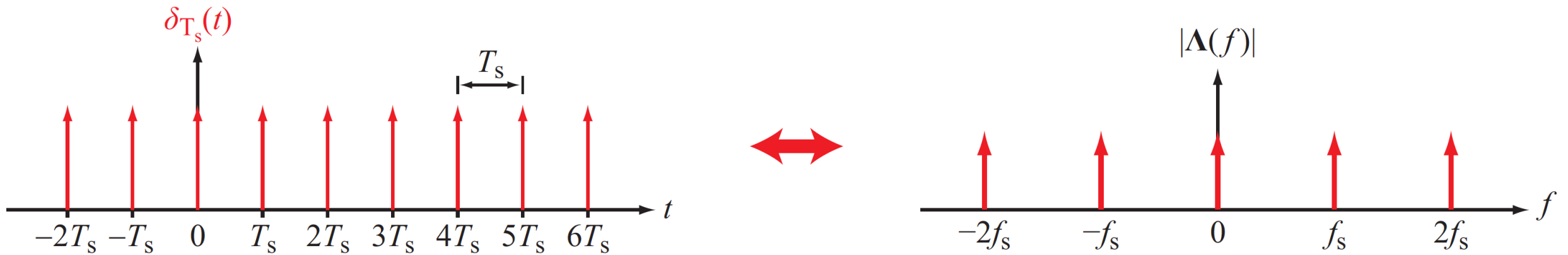
Sampling Theorem

- Let $x(t)$ be a real-valued, continuous-time, lowpass signal *bandlimited* to a maximum frequency of B Hz.
- Let $x[n] = x(nT_s)$ be the sequence of numbers obtained by *sampling* $x(t)$ at a sampling rate of f_s samples per second, that is, every $T_s = 1/f_s$ seconds.
- Then $x(t)$ can be *uniquely* reconstructed from its samples $x[n]$ if and only if $f_s > 2B$. The sampling rate must *exceed* double the bandwidth.



Recap: Fourier Series of the Pulse Train

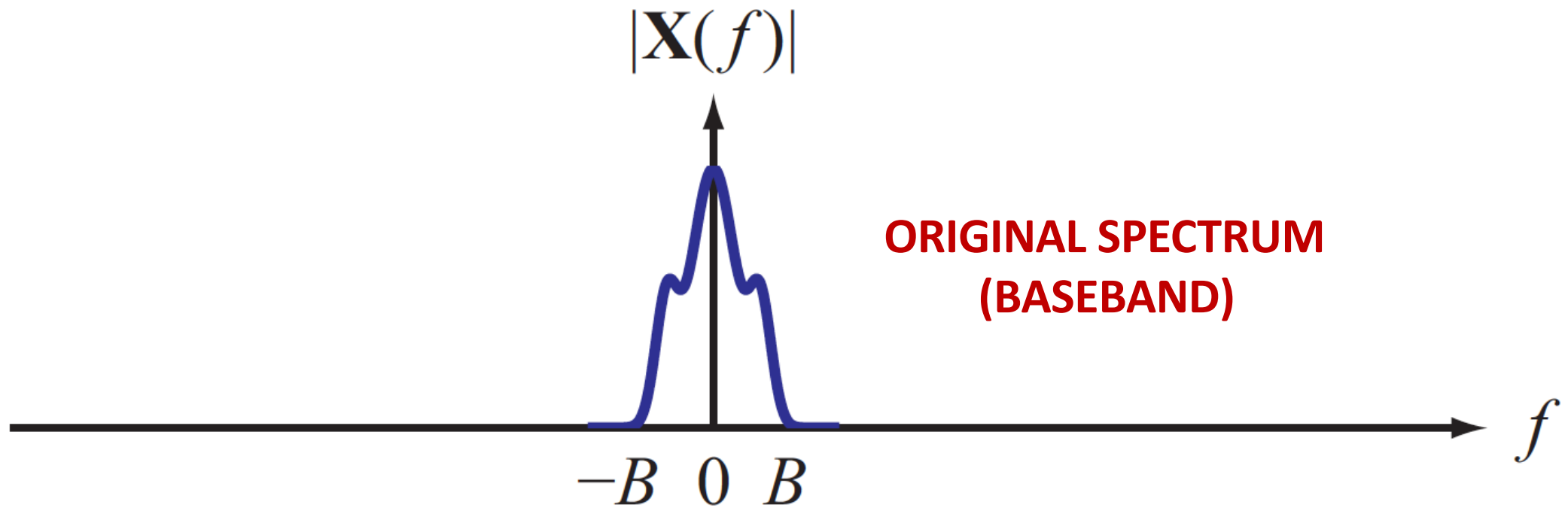
$$\Lambda(f) = \mathcal{F}[\delta_{T_s}(t)] = f_s \sum_{n=-\infty}^{\infty} \delta(f - nf_s)$$



FASCINATING:
**THE FOURIER TRANSFORM OF THE PULSE TRAIN IS A PULSE TRAIN,
IN THE FREQUENCY DOMAIN**



Fourier Transform of the Signal

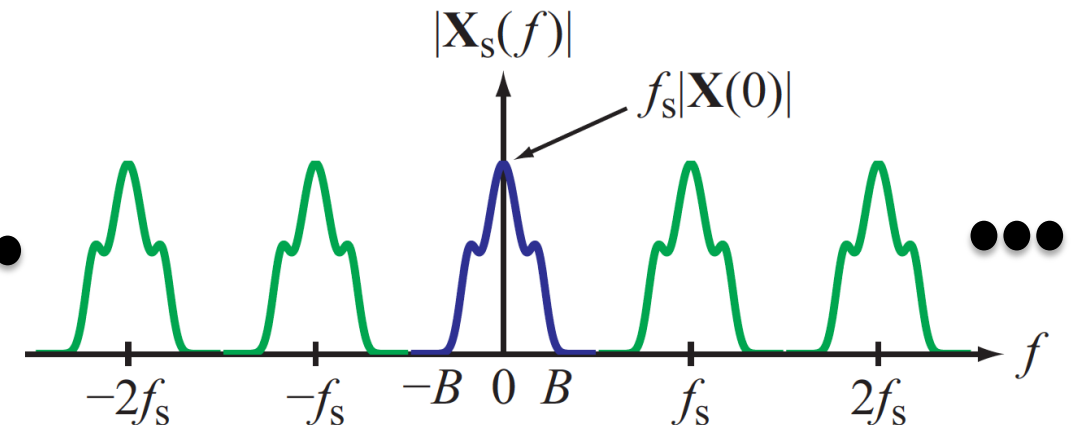
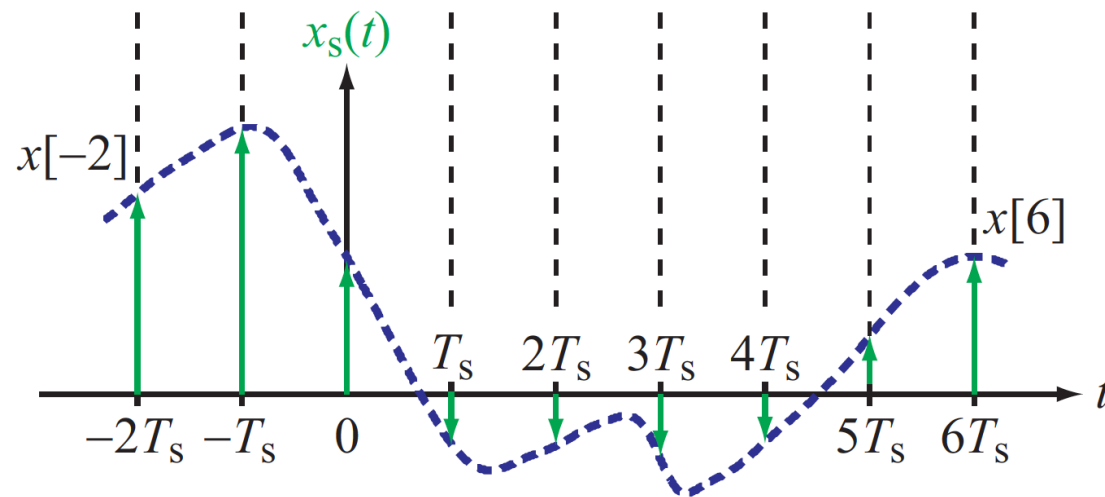




Fourier Series of the Sampled Signal

CONVOLVE

$$\mathbf{X}_s(f) = \mathbf{X}(f) * f_s \sum_{n=-\infty}^{\infty} \delta(f - nf_s) = f_s \sum_{n=-\infty}^{\infty} \mathbf{X}(f - nf_s)$$

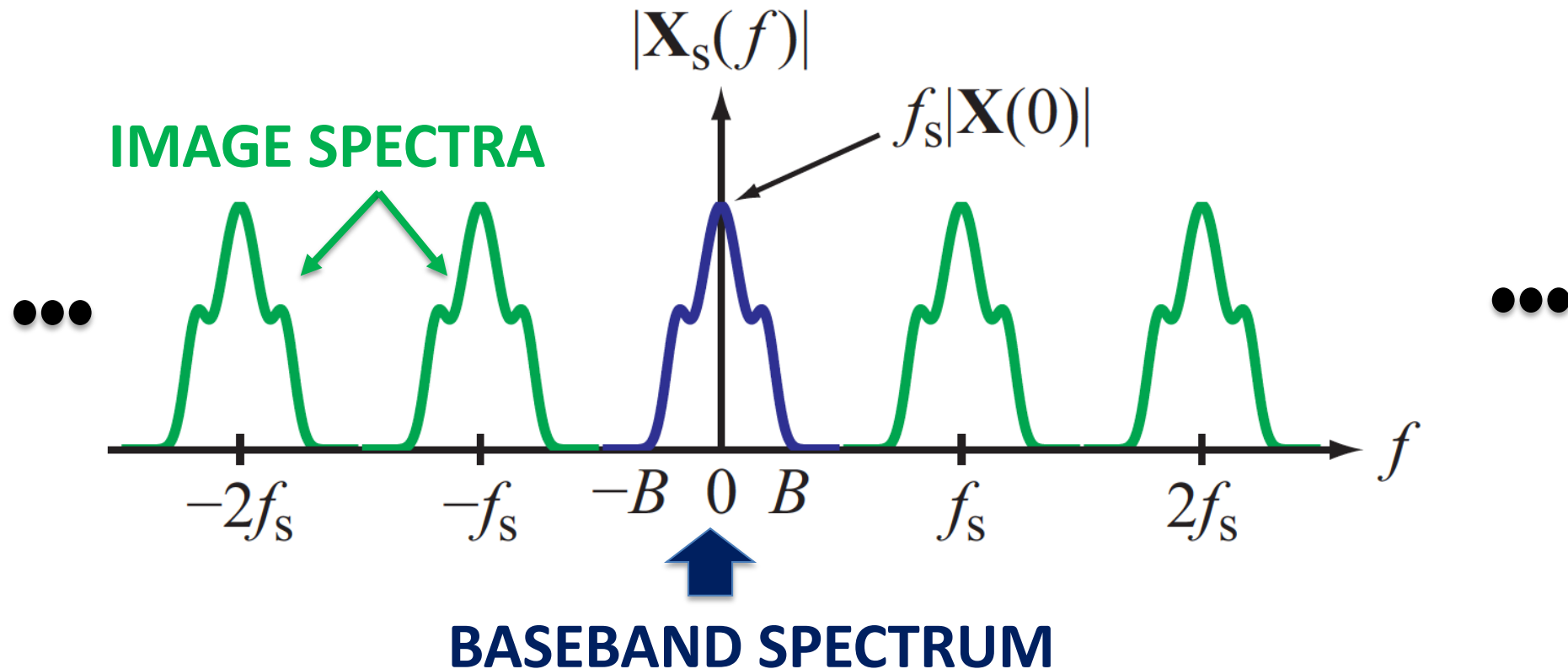


WE CAN NOW EXAMINE THE EFFECT OF VARYING THE SAMPLING FREQUENCY



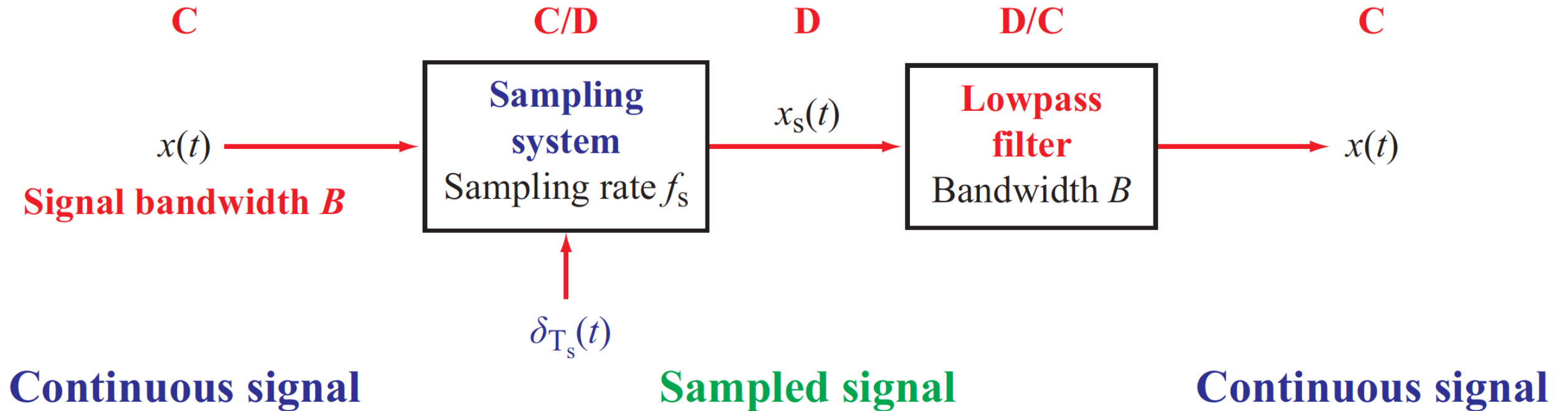
Fourier Series of the Sampled Signal

$$\mathbf{X}_s(f) = \mathbf{X}(f) * f_s \sum_{n=-\infty}^{\infty} \delta(f - nf_s) = f_s \sum_{n=-\infty}^{\infty} \mathbf{X}(f - nf_s)$$





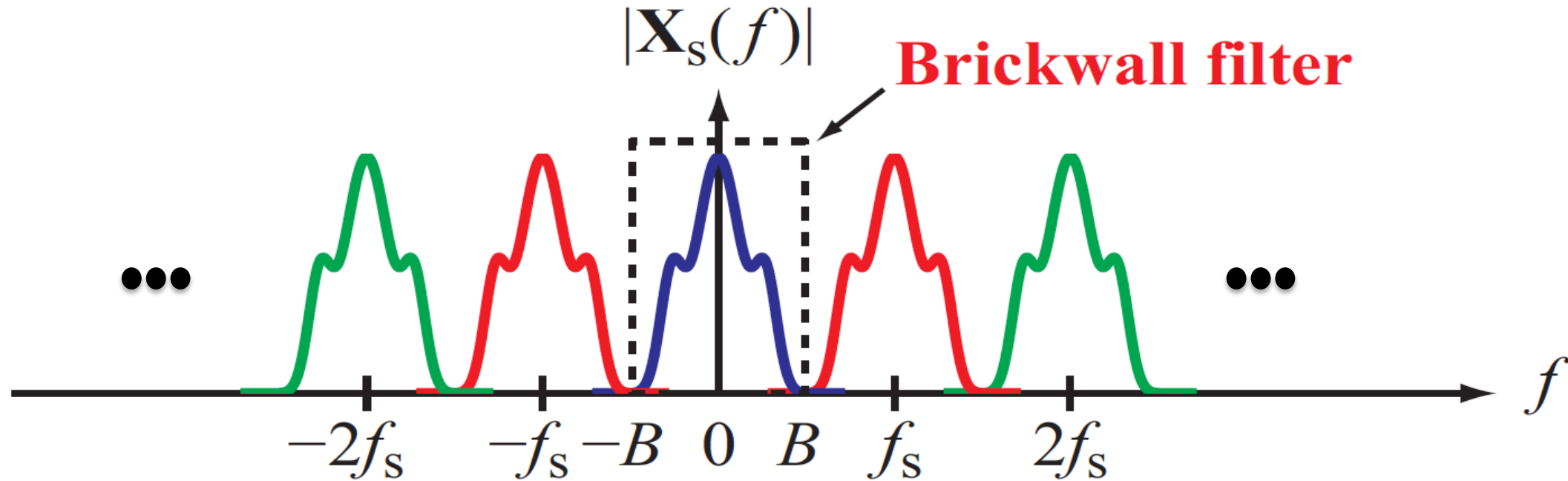
Recovering the Signal



C: CONTINUOUS TIME
D: DISCRETE TIME



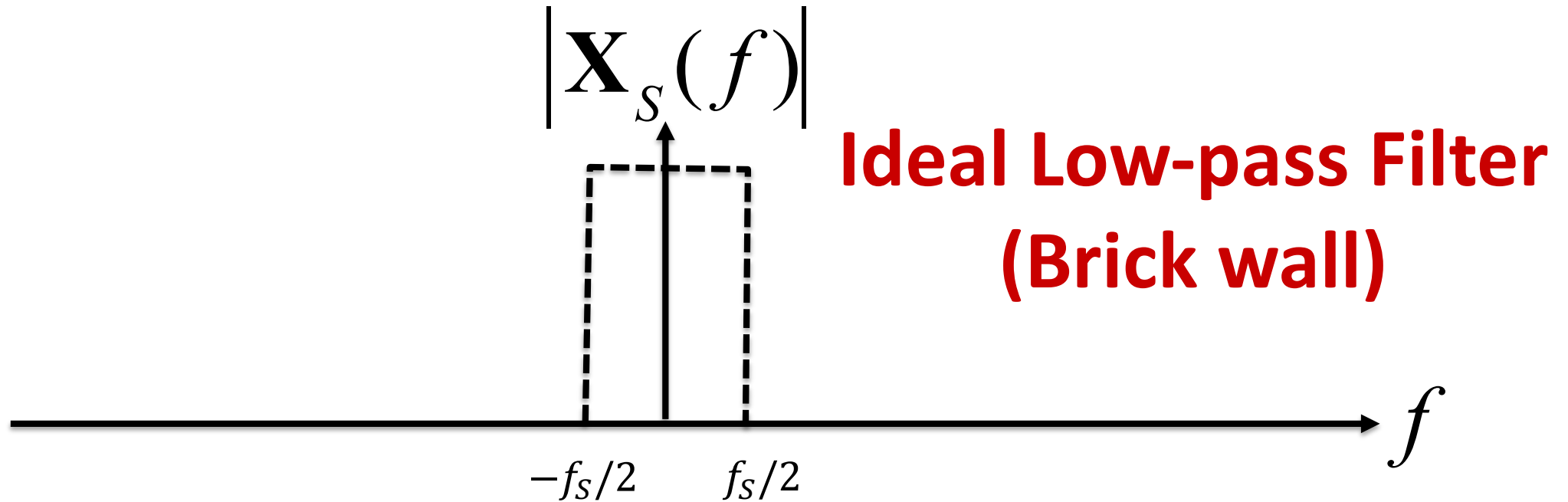
Isolating the Baseband Spectrum



(c) Nyquist sampled ($f_s = 2B$)



Recovering the Signal



The Inverse Fourier transform of the ideal LP filter is the sinc function:

$$h(t) = \frac{1}{f_s} \frac{\sin(\pi f_s t)}{\pi t} = \frac{\sin(\pi f_s t)}{\pi f_s t}$$



Recovering the Signal

To recover the signal $x(t)$, we convolve the sampled signal with $h(t)$

$$x(t) = x_s(t) * h(t)$$

$$x_s(t) = \sum_{n=-\infty}^{\infty} x(nT_s) \delta(t - nT_s)$$



Recovering the Signal

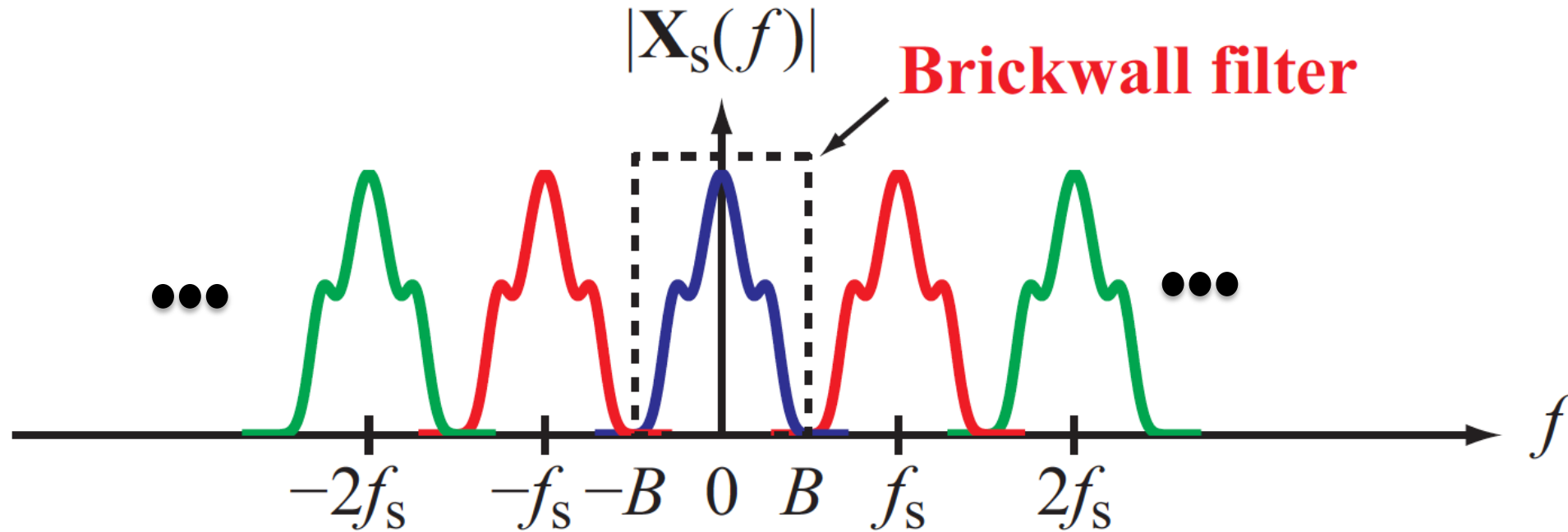
CONVOLVE THE SAMPLED SIGNAL WITH THE LOW-PASS FILTER:

$$\begin{aligned}x(t) &= x_s(t) * h(t) = \sum_{n=-\infty}^{\infty} x(nT_s) \delta(t - nT_s) * h(t) \\&= \sum_{n=-\infty}^{\infty} x(nT_s) h(t - nT_s) \\&= \sum_{n=-\infty}^{\infty} x(nT_s) \frac{\sin(\pi f_s(t - nT_s))}{\pi f_s(t - nT_s)}\end{aligned}$$

THIS IS SHANNON'S INTERPOLATION (CONNECT THE DOTS) FORMULA FOR RECOVERING THE SIGNAL DIRECTLY FROM ITS SAMPLES



Practical Issue !

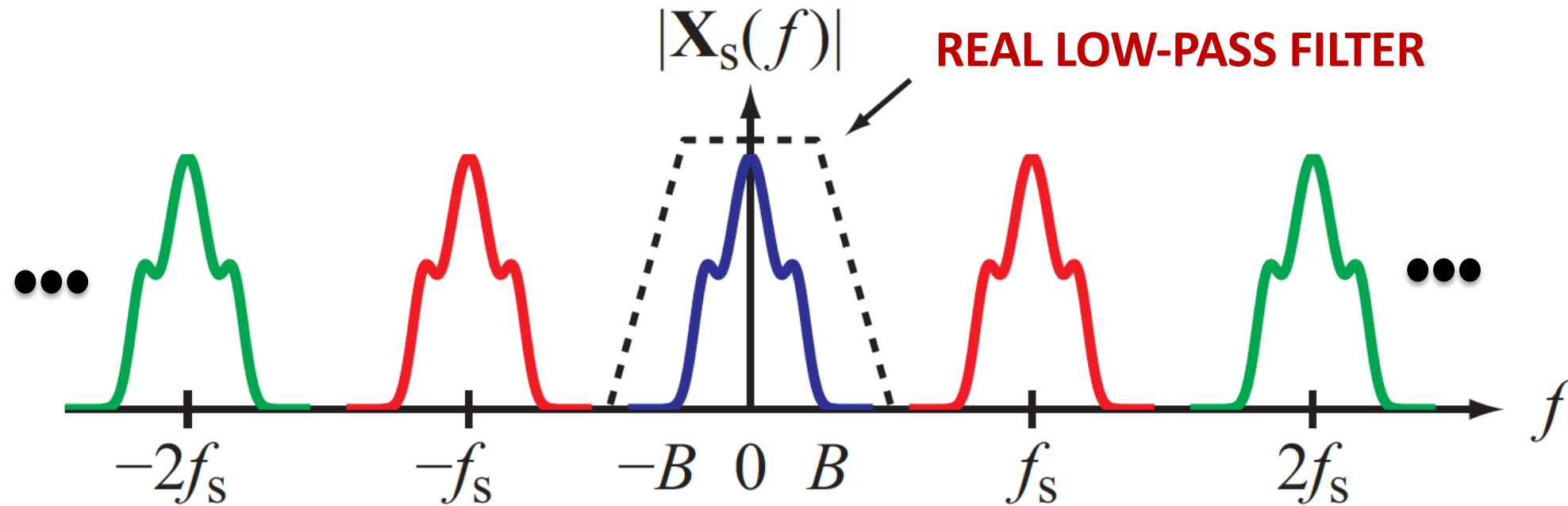


(c) Nyquist sampled ($f_s = 2B$)

WE CAN RECOVER THE SIGNAL PERFECTLY, IN THEORY
ISSUE: NEEDS AN IDEAL LOW-PASS “BRICK WALL” FILTER
UNFORTUNATELY, THAT IS NOT IMPLEMENTABLE !



Making it Practical

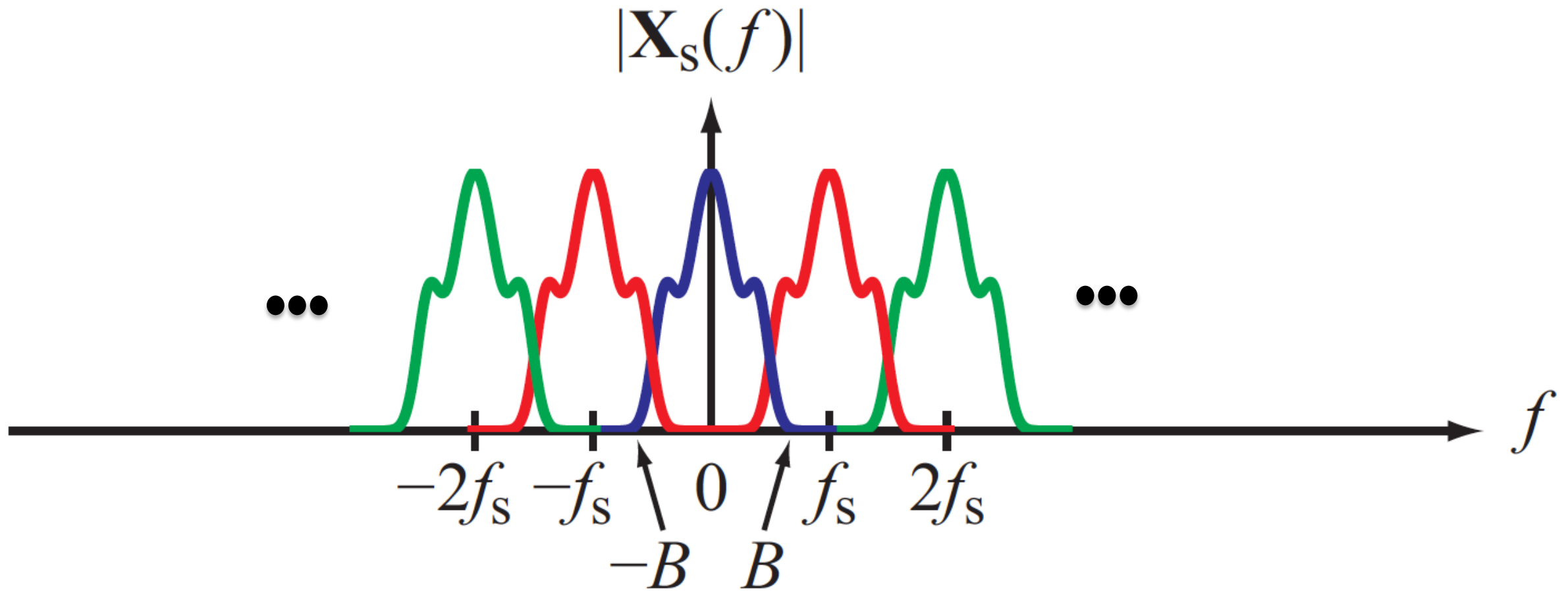


(d) Oversampled ($f_s > 2B$)

BASIC IDEA: ADD BLANK SPACES BETWEEN THE SPECTRA TO MAKE IT IMPLEMENTABLE WITH A REAL LOW-PASS FILTER



Under Sampling



THE RECOVERED SIGNAL IS DISTORTED



Example 1

A 200Hz sinusoidal signal is sampled at 500 Hz
(more than the Nyquist Rate)

$$x(t) = \cos(400\pi t)$$

$$x_s(t) = \sum_{n=-\infty}^{\infty} x(t) \delta(t - nT_s) \quad \text{SAMPLED SIGNAL}$$

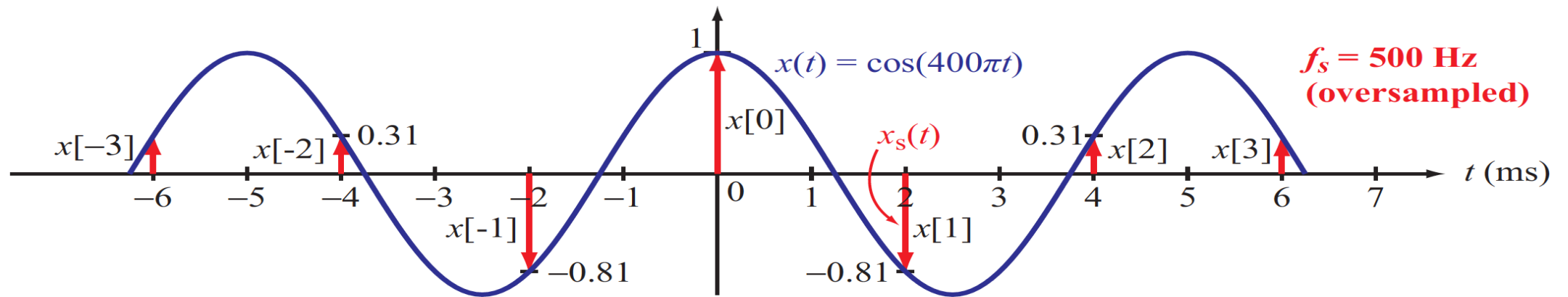
$$= \sum_{n=-\infty}^{\infty} \cos(400\pi t) \delta(t - 2 \times 10^{-3}n)$$

$$= \sum_{n=-\infty}^{\infty} \cos(0.8\pi n) \delta(t - 2 \times 10^{-3}n)$$

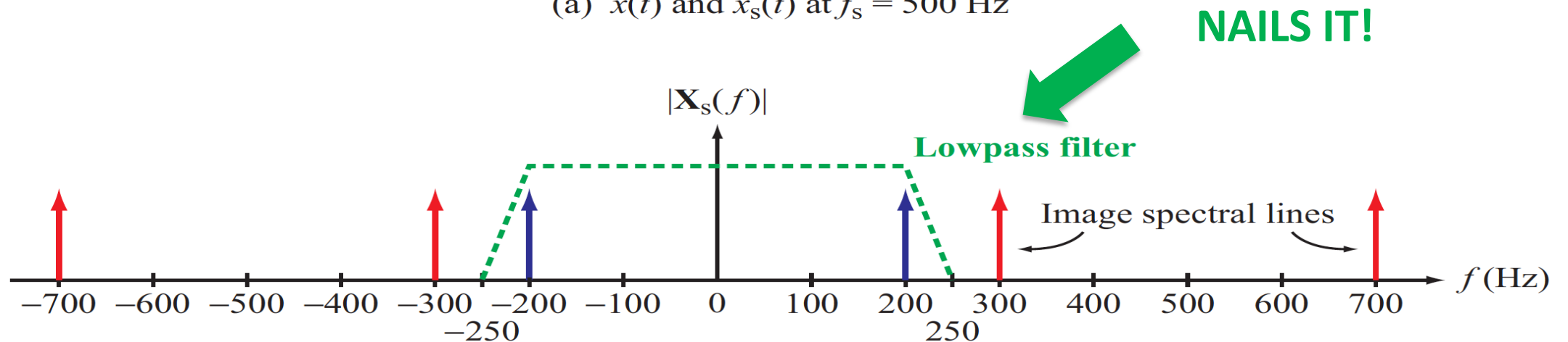


Example 1: Above Nyquist

A 200Hz sinusoidal signal is sampled at 500 Hz
(more than the Nyquist Rate)



(a) $x(t)$ and $x_s(t)$ at $f_s = 500$ Hz

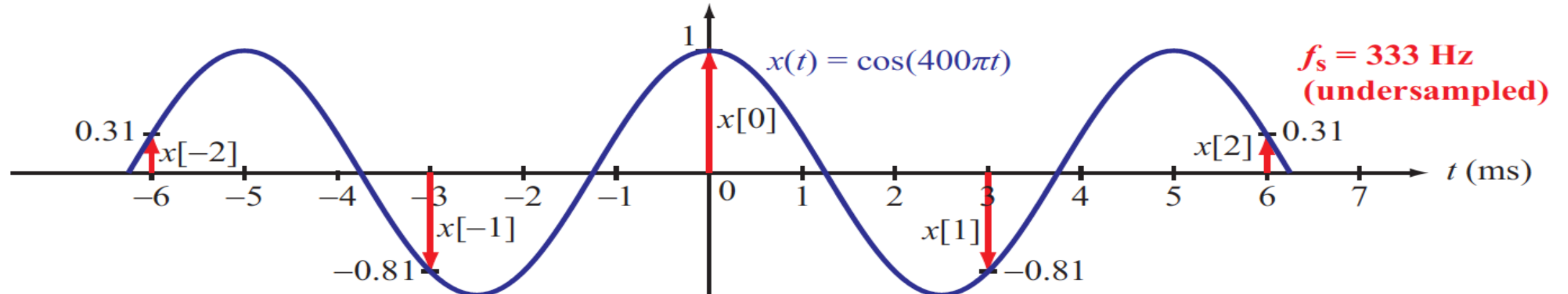


(b) Spectrum of $\mathbf{X}_s(f)$ [blue = spectrum of $x(t)$; red = image spectra]



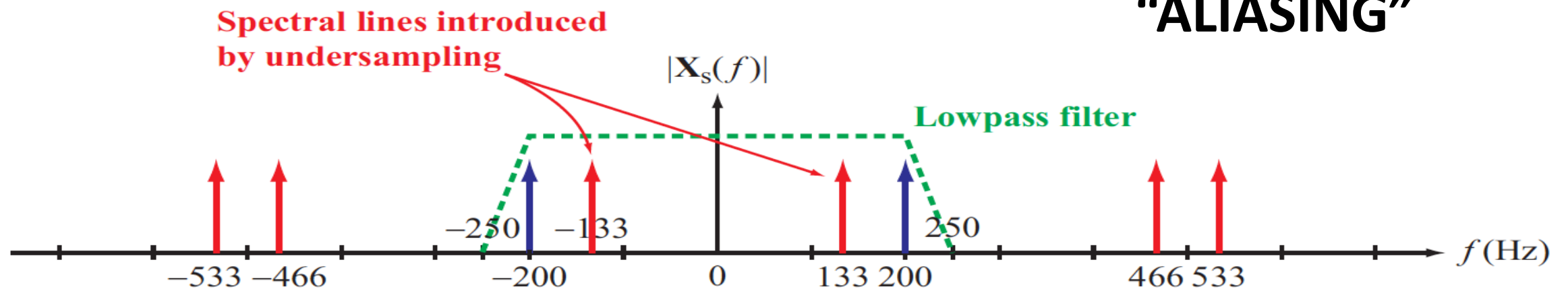
Example 2: Below Nyquist

A 200Hz sinusoidal signal is sampled at 333 Hz



(a) $x(t)$ and $x_s(t)$ at $f_s = 333 \text{ Hz}$

“ALIASING”



(b) Spectrum of $X_s(f)$ [blue = spectrum of $x(t)$; red = image spectra]



Aliasing

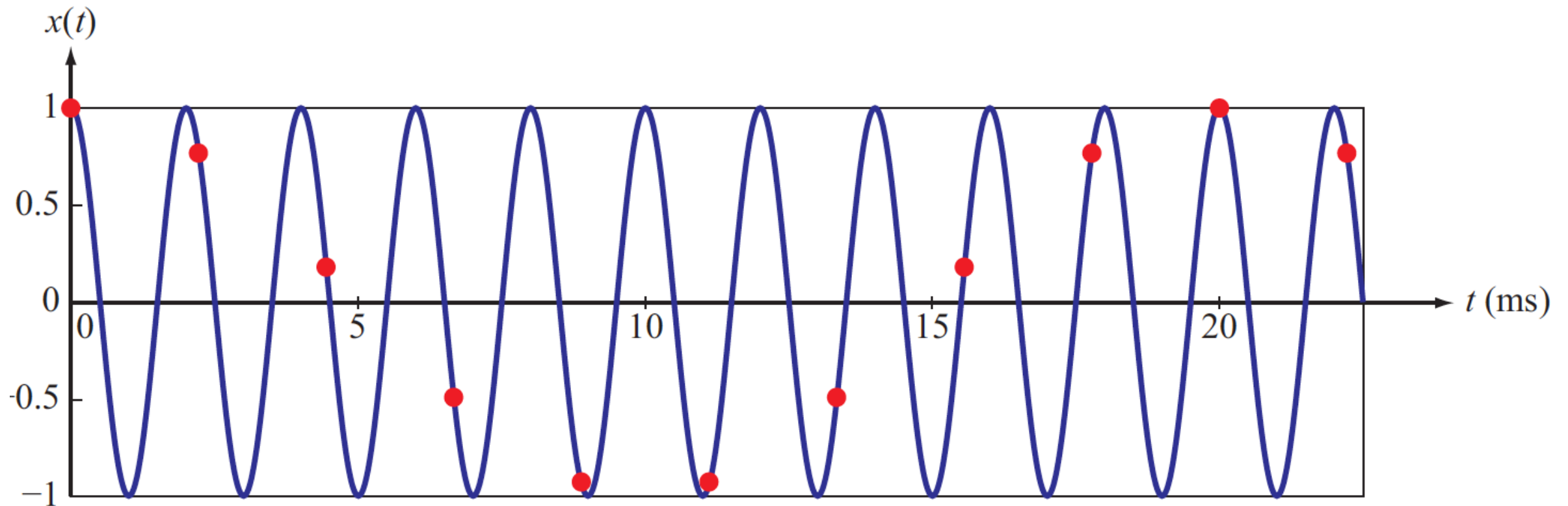
The reason why we use the word “aliasing”

- When we sample a signal at a rate below the Nyquist rate, the high-frequency signals don't just disappear!
- The samples from the high frequencies show up as fake “reflections” or aliases in the baseband
- They can manifest in confusing ways in the time domain and are hard to clean up since they are inextricably mixed in with the baseband
- It is best to avoid aliasing in the first place



Aliasing can be Confusing

500Hz sinusoid sampled at 450Hz looks like a (fake) 50Hz signal!



FOLLOW THE RED DOTS



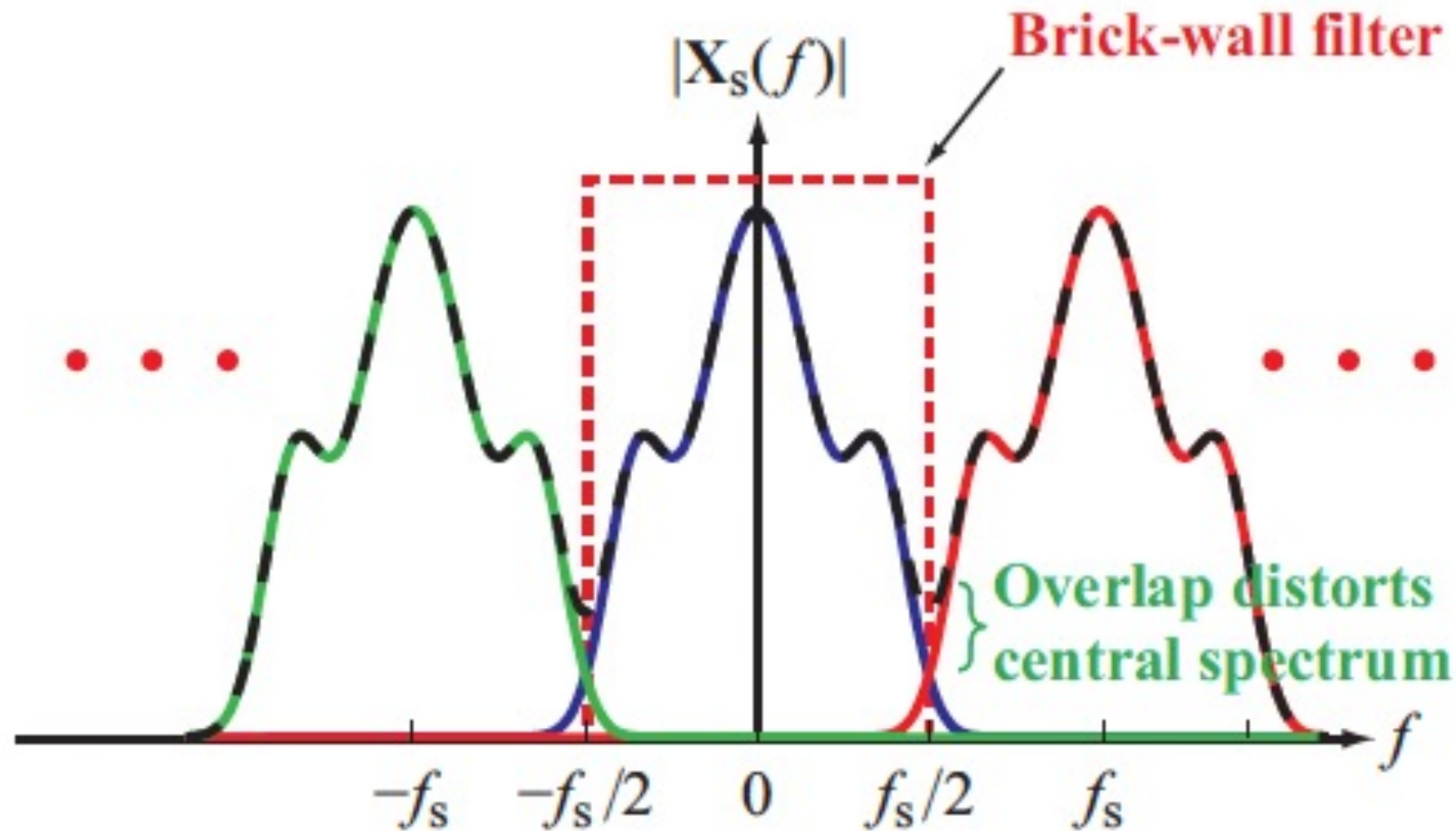
Aliasing Example



<https://youtu.be/ByTsISFXUoY>



Aliasing in Real Signals

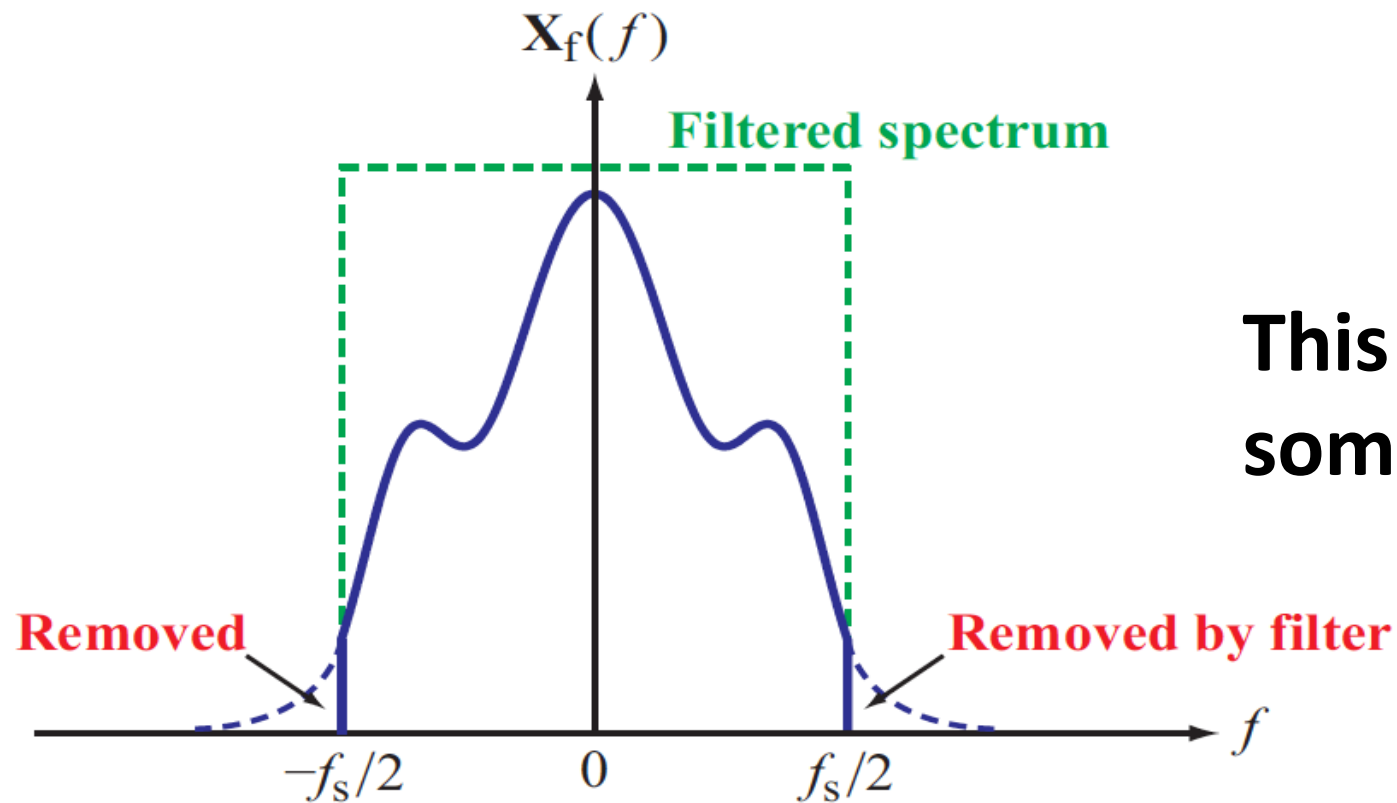
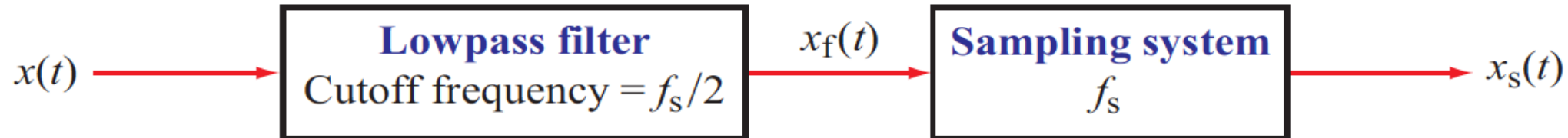


If we examine the spectrum of the sampled signal, we see that there is distortion from the “tails” of the shifted spectra



The Anti-aliasing Filter

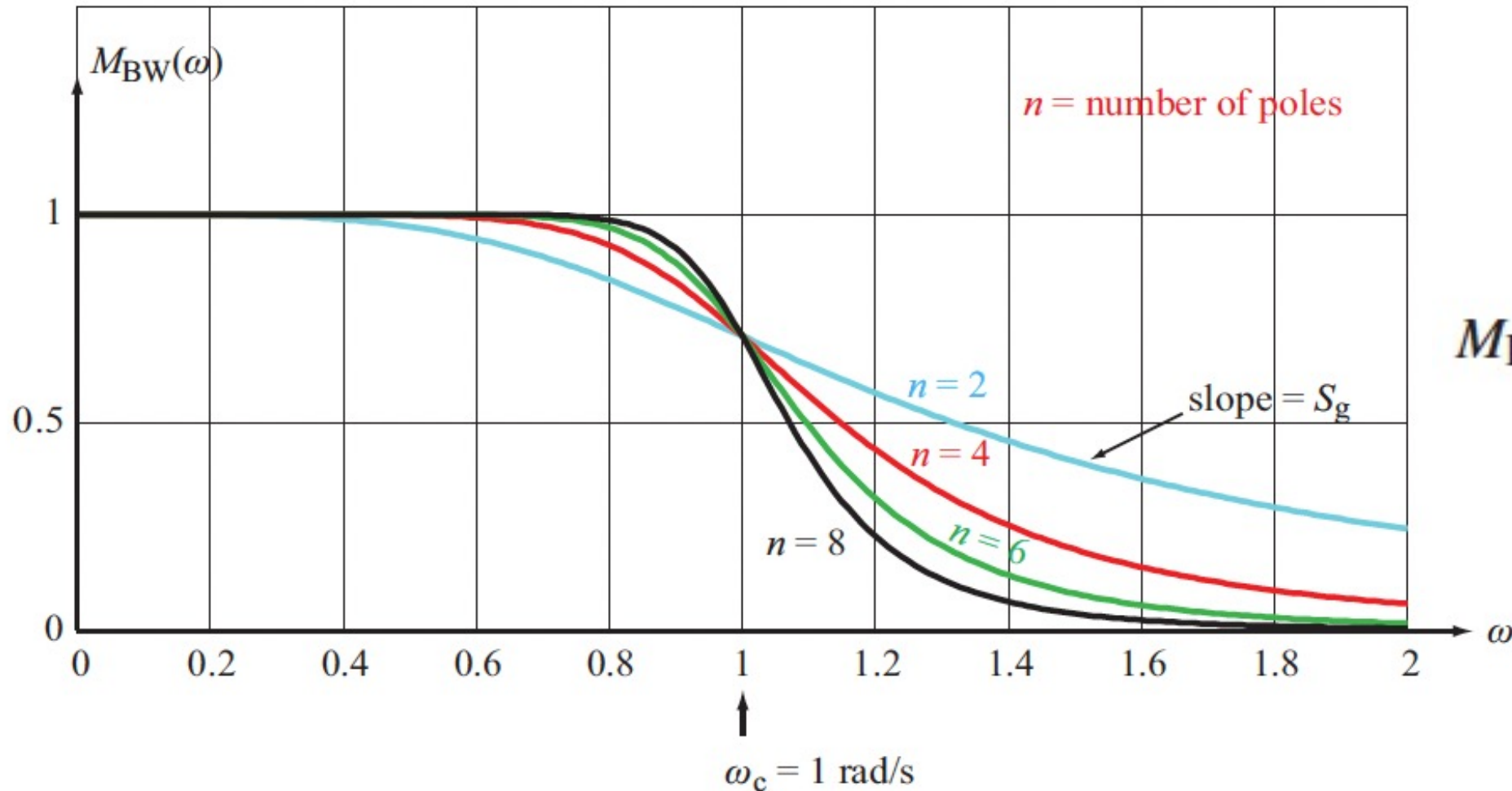
We can force the sampled signal to be bandlimited



This necessarily entails
some loss of content



A More Realistic Filter: Butterworth



$$M_{LP}(f) = \frac{1}{\sqrt{1 + (f/f_c)^{2n}}}$$

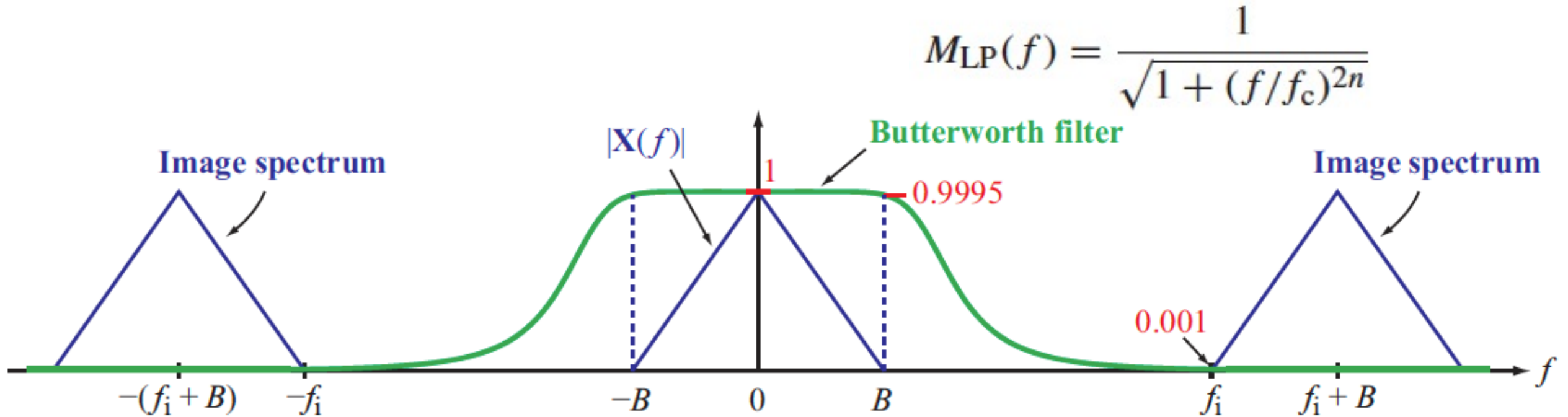
Butterworth Filters provide a steeper roll-off compared to the RC circuit ($n = 1$)

By increasing the order of the filter (n), we can get better filtering

BIBO stable, causal, realizable with op-amps, R's, C's



More Realistic Filter Example



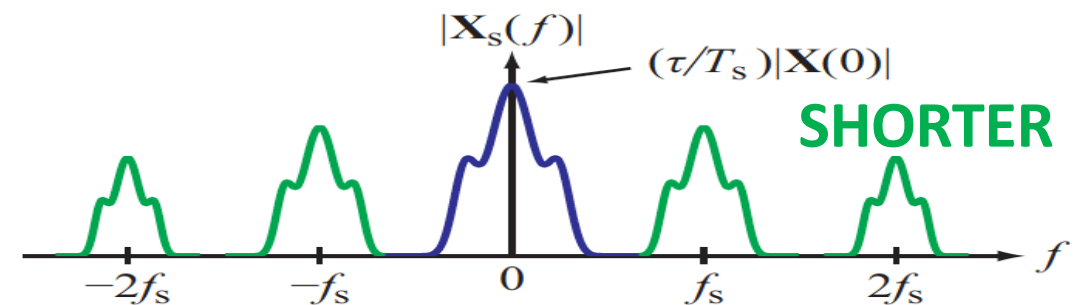
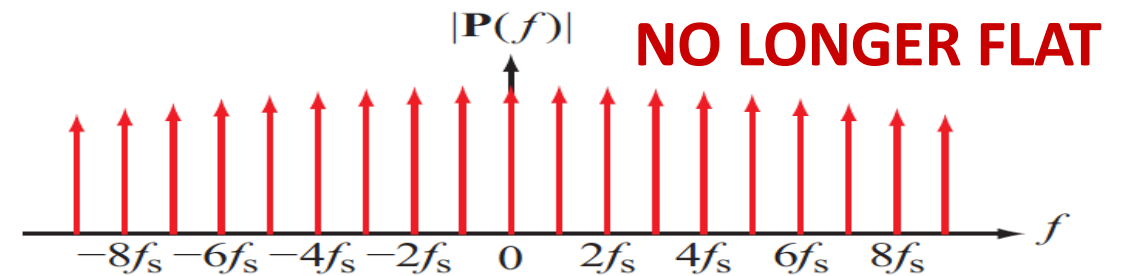
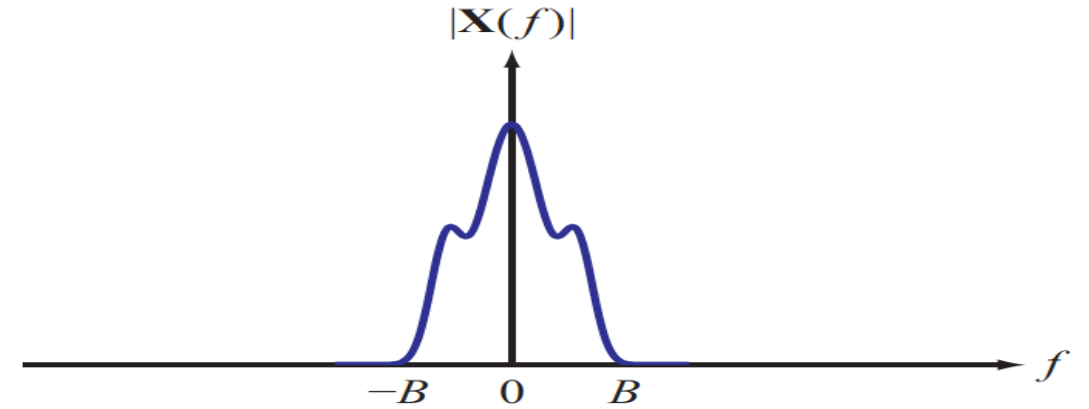
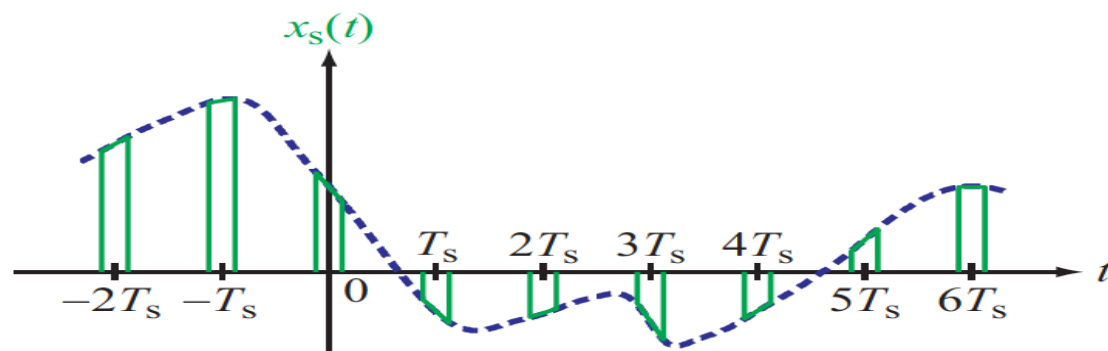
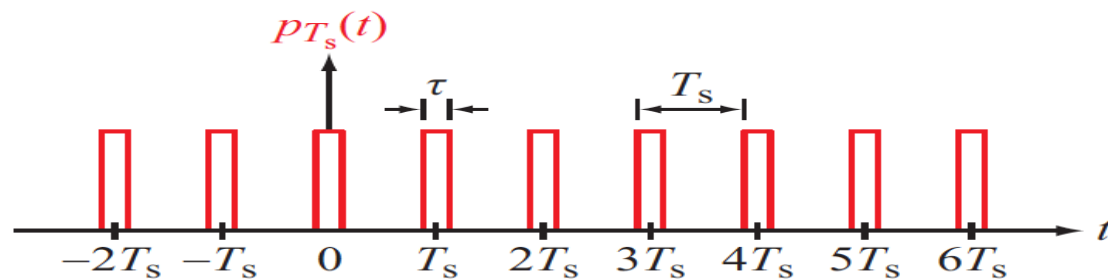
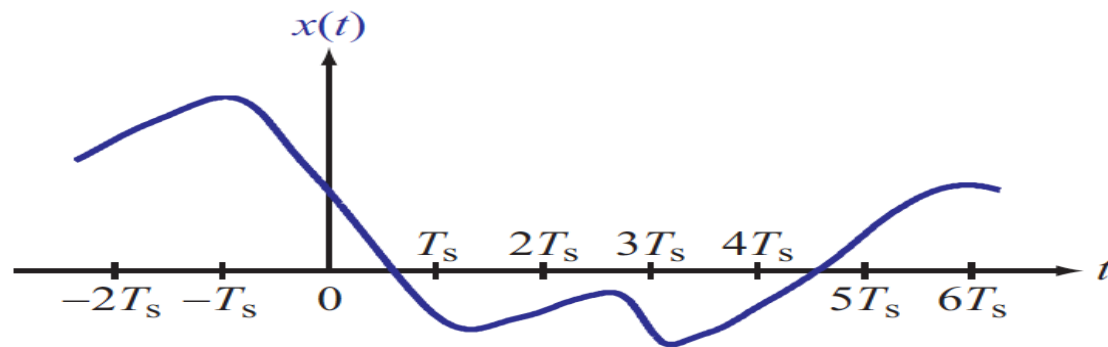
In practice, we specify an acceptable level for distortion, for example, requiring that the filter response is $\leq 0.1\%$ of the peak

- For a given filter, this defines the minimum sampling frequency. It can be much greater than $2B$, and for this example, it is $9B$
- The tolerance level can also be used to select a filter with a sufficiently rapid roll-off, for example, Butterworth with $n = 5$



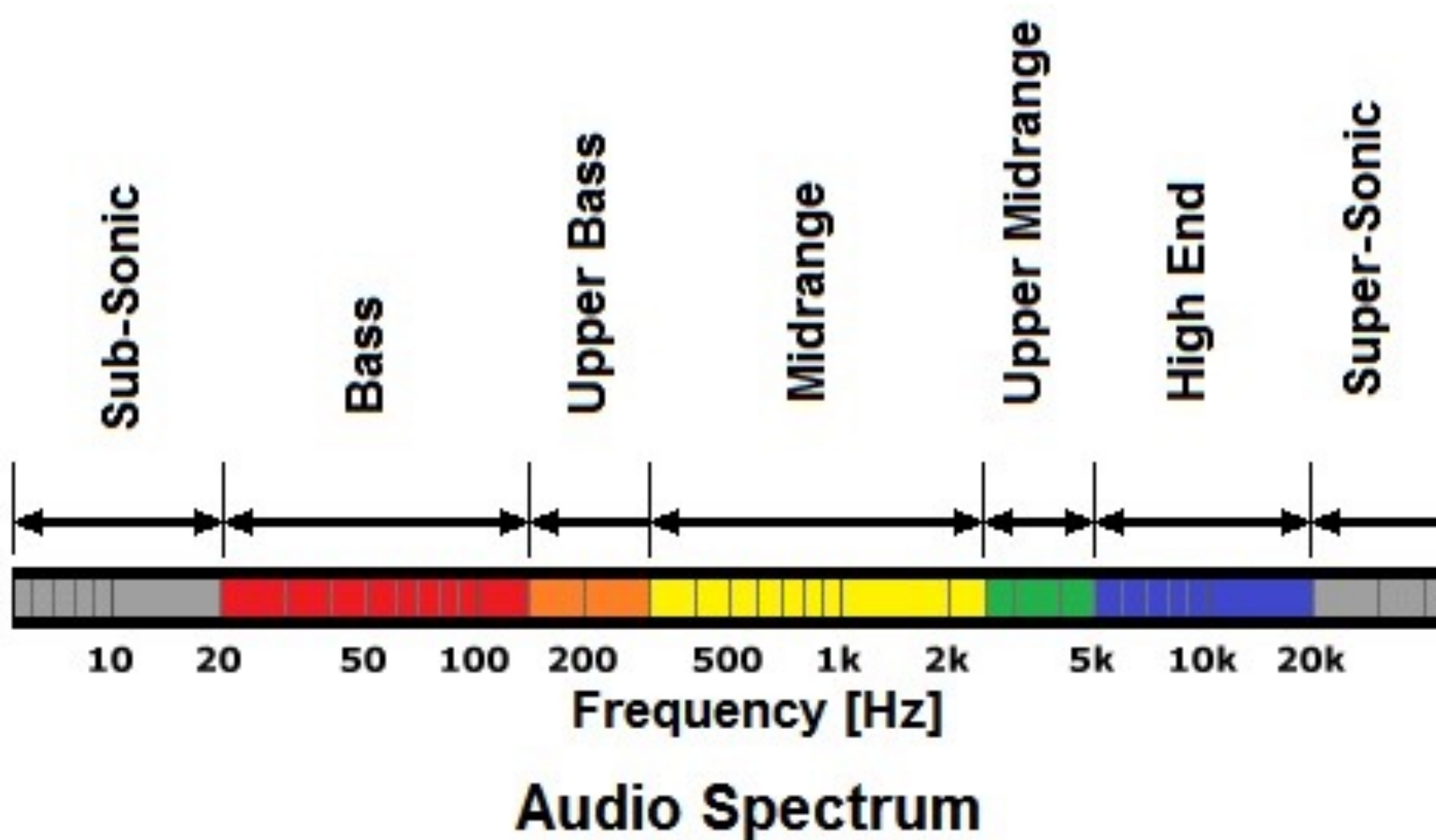
Another Practical Matter

Ideal impulses are unachievable, so we use narrow pulses instead





Practical Example: CD Quality Audio

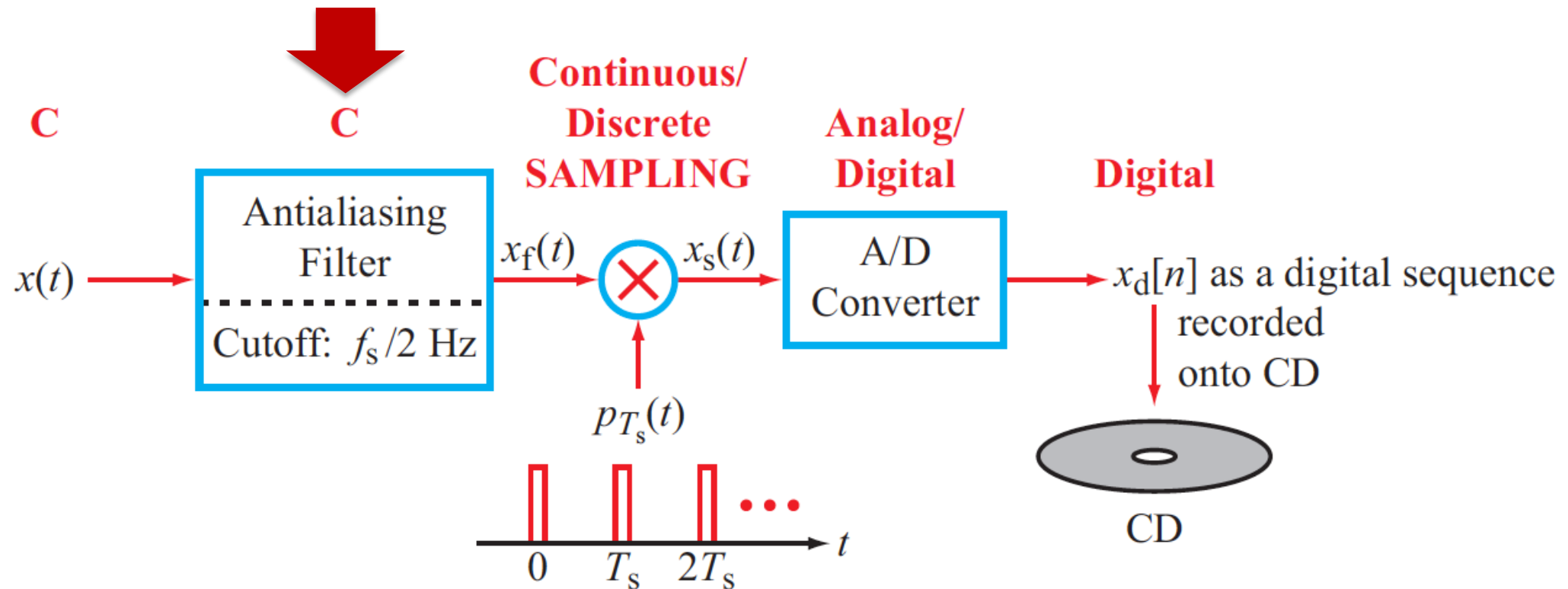


Audio systems are designed to handle signals in the 20Hz – 20kHz range



Practical Example: CD Audio

We add an anti-aliasing filter



Expected sound frequencies: 20 Hz – 20 kHz

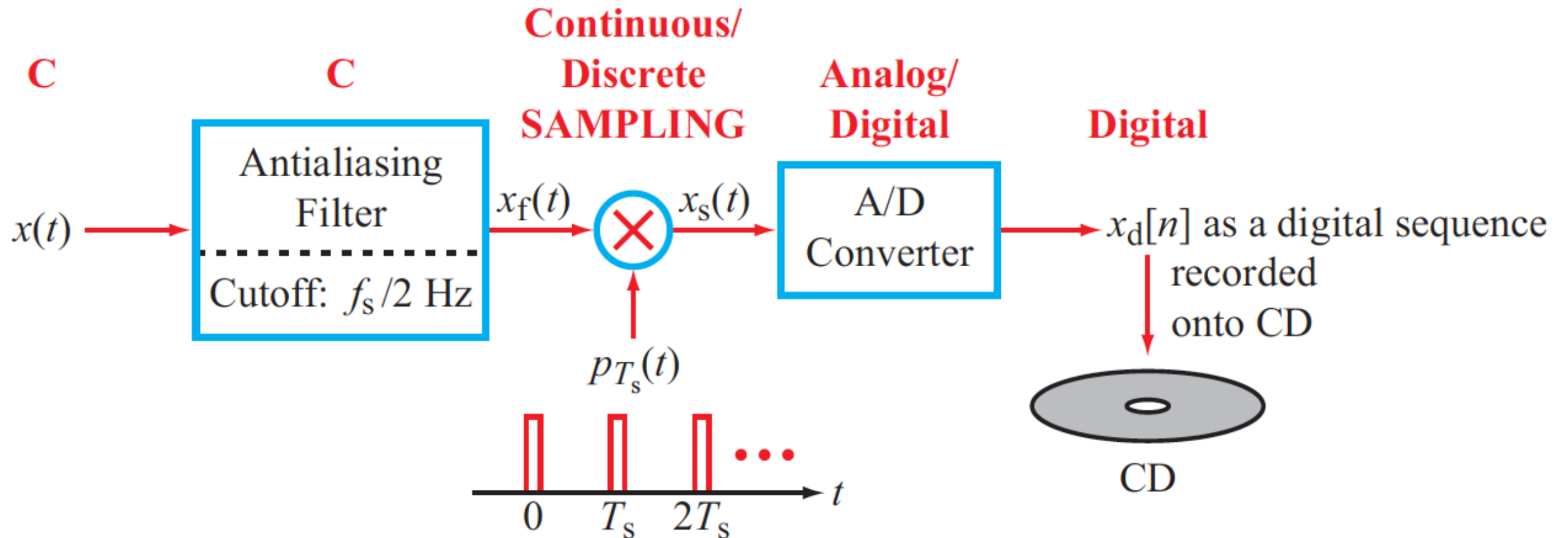
Chosen “padded” cutoff for the filter = 22.05 kHz

$\Rightarrow$ Required Sampling Rate = 44.1 kHz



Practical Example: CD Audio

We use two copies of this system (left, right)

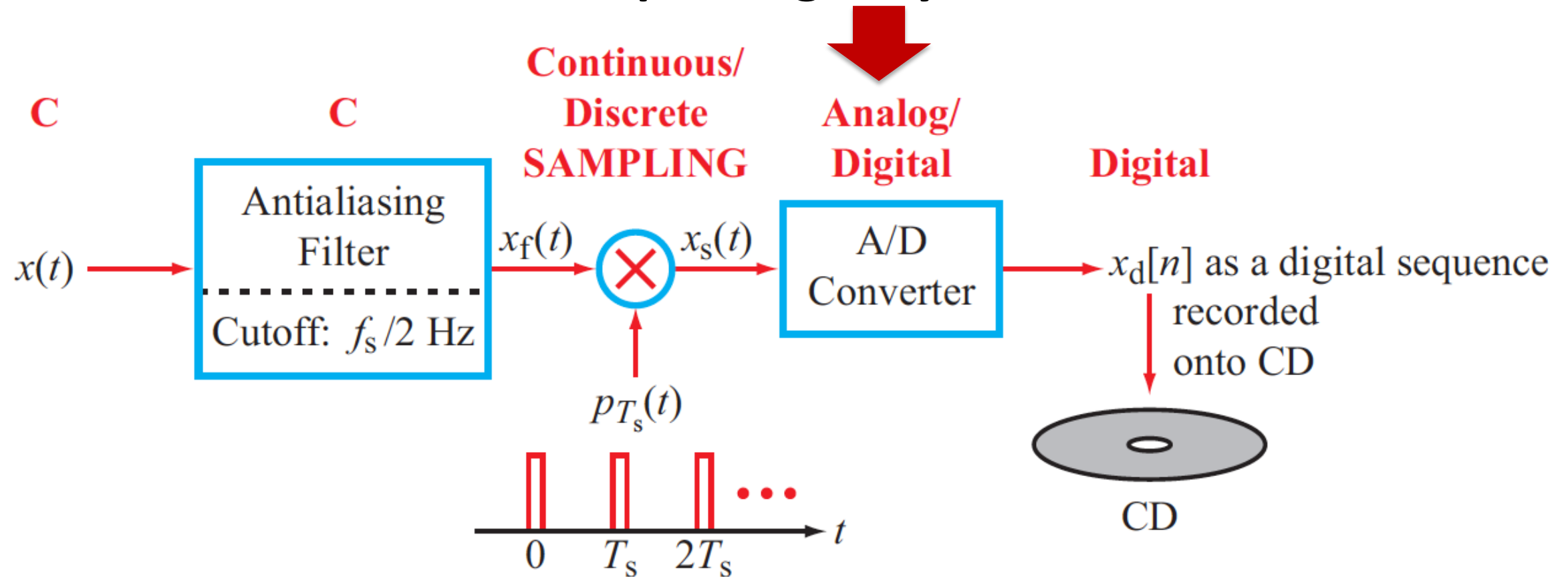


For recording stereophonic sounds (for left and right ear)



Practical Example: CD Audio

We store the samples digitally



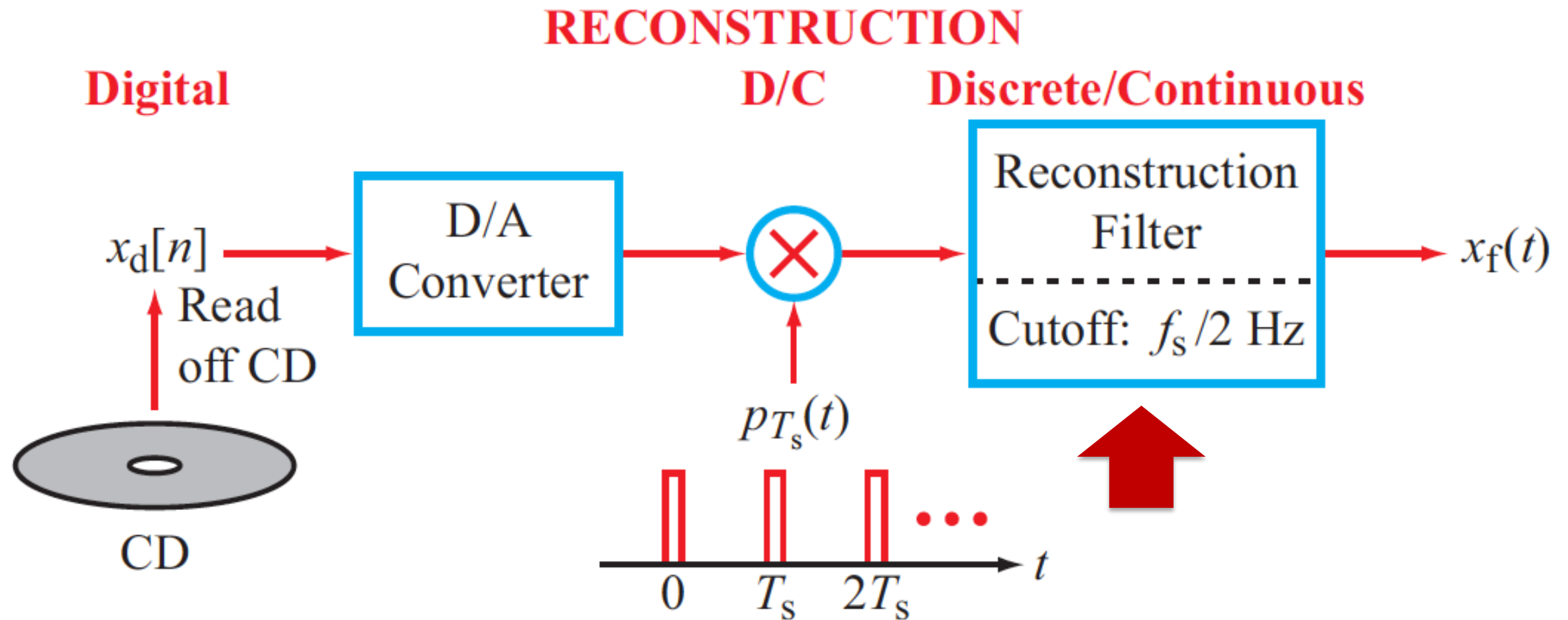
Precision = 16 bits / sample

Two stereo channels: Left and Right



Practical Example: CD Audio

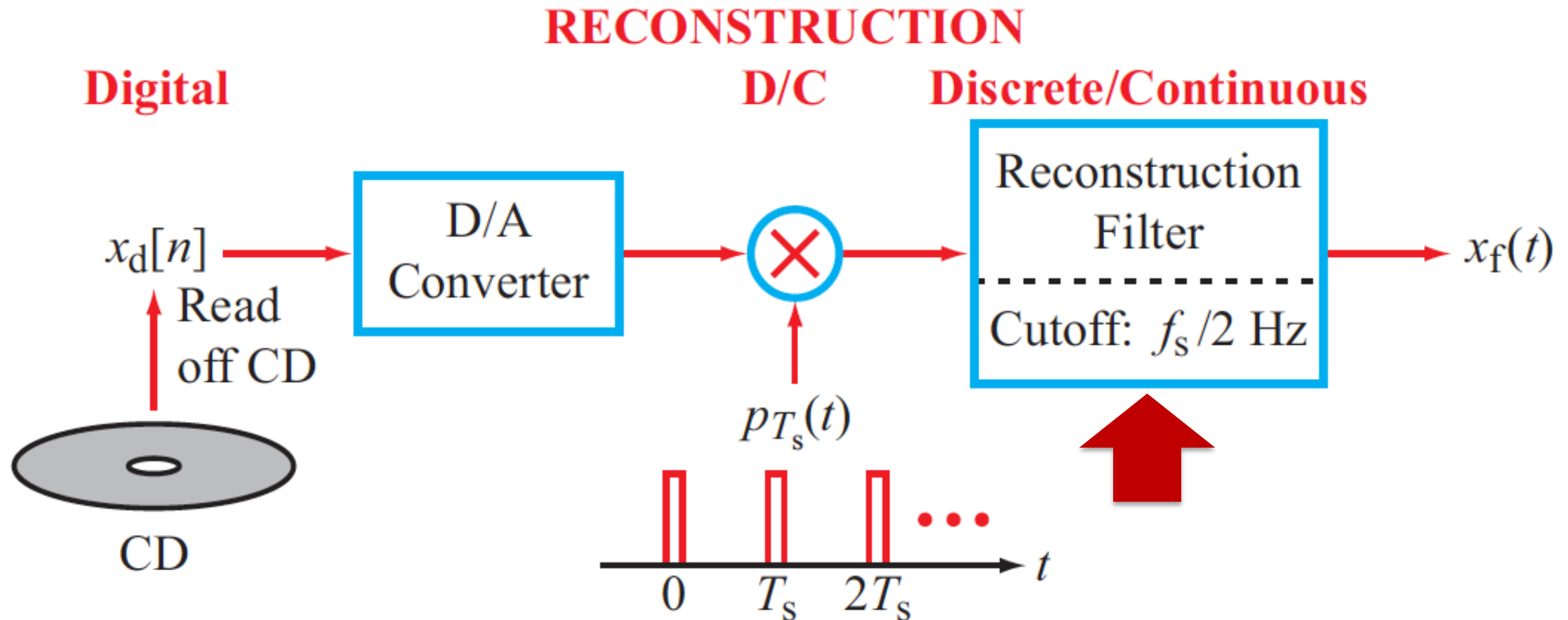
The reconstruction filter is the same as the anti-aliasing filter





Practical Example: CD Audio

Some audio systems choose to “oversample” the signal in the digital domain to make the filter design easier (gentler roll-offs), but deceptively imply in the marketing materials that it is a better product (it is not!)





Summary

Sampling Theorem

- Let $x(t)$ be a real-valued, continuous-time, lowpass signal *bandlimited* to a maximum frequency of B Hz.
- Let $x[n] = x(nT_s)$ be the sequence of numbers obtained by *sampling* $x(t)$ at a sampling rate of f_s samples per second, that is, every $T_s = 1/f_s$ seconds.
- Then $x(t)$ can be *uniquely* reconstructed from its samples $x[n]$ if and only if $f_s > 2B$. The sampling rate must exceed double the bandwidth.

This theorem is our "bridge" to the world of digital signal processing



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